



Powers and roots

A POWER IS THE NUMBER OF TIMES A NUMBER IS MULTIPLIED BY ITSELF. THE ROOT OF A NUMBER IS A NUMBER THAT, MULTIPLIED BY ITSELF, EQUALS THE ORIGINAL NUMBER.

SEE ALSO

◀ 18–21 Multiplication

◀ 22–25 Division

Standard form 42–43 ▶

Using a calculator 72–73 ▶

Introducing powers

A power is the number of times a number is multiplied by itself. This is indicated as a smaller number positioned to the right above the number. Multiplying a number by itself once is described as “squaring” the number; multiplying a number by itself twice is described as “cubing” the number.

5⁴

this is the power, which shows how many times to multiply the number (5⁴ means 5 × 5 × 5 × 5)

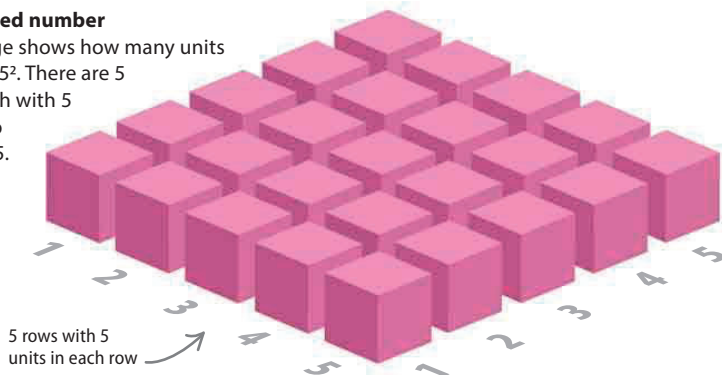
this is the number that the power relates to

$$5 \times 5 = 5^2 = 25$$

this is the power; 5² is called “5 squared”

▷ Squared number

This image shows how many units make up 5². There are 5 rows, each with 5 units—so 5 × 5 = 25.



$$5 \times 5 \times 5 = 5^3 = 125$$

this is the power; 5³ is called “5 cubed”

△ The cube of a number

Multiplying a number by itself twice gives its cube. The power for a cube number is ³, for example 5³, which means there are 3 number 5's being multiplied: 5 × 5 × 5.

5 vertical rows

▷ Cubed number

This image shows how many units make up 5³. There are 5 horizontal rows and 5 vertical rows, each with 5 units in each one, so 5 × 5 × 5 = 125.

